Pandas Quick Cheat Sheet

Setup and inspection

```
import pandas as pd
df = pd.read_csv("servers.csv")
print(df.head())             # first five rows
print(df.info())             # columns, types, missing data
print(df.describe())         # numeric summary
print(df.shape, df.columns)  # dimensions and labels
```

Create and select

```
s = pd.Series([10, 20, 30])
df = pd.DataFrame({"host": ["a", "b"], "cpu": [45, 91]})
df["cpu"]                  # one column (Series)
df[["host", "cpu"]]        # multiple columns (DataFrame)
df.loc[0, "cpu"]           # label-based selection
df.iloc[0, 1]              # position-based selection
```

Filter and sort

```
high = df[df["cpu"] > 80]
subset = df[(df["cpu"] > 80) & (df["host"] != "b")]
ordered = df.sort_values("cpu", ascending=False)
```

Summarise and group

```
df["cpu"].mean()
df["host"].value_counts()
df.groupby("region")["cpu"].mean()
df.groupby("status").agg(count=("host", "size"),
                         avg_cpu=("cpu", "mean"))
```

Transform, clean, export

```
df["risk"] = df["cpu"].apply(lambda x: "High" if x > 80 else "OK")
df["cpu"] = pd.to_numeric(df["cpu"], errors="coerce")
df = df.dropna(subset=["cpu"])
df.to_csv("servers_report.csv", index=False)
```

Charts and common pitfalls

```
import matplotlib.pyplot as plt
df["status"].value_counts().plot(kind="bar")
plt.tight_layout(); plt.savefig("status_counts.png")
# Use & / | with parentheses, not and / or on Series.
# Check exact column names and data types before filtering.
```